

Sec. VI-E, (98)–(108): the onset under a perturbation, and the threshold

drop-in main text; the measured sizes live in Sec. VIII-E

Because the measured angular rate $y(t)$ can be regarded as the nominal response perturbed by the unmodelled forcing ρ , the minimiser selects $\hat{t}_c$ by minimising the two-segment cost over the record. Both segments enter, and both share the pre-onset level C :

$$J(t_c) = \int_{t_0}^{t_c} (y(t) - C)^2 dt + \int_{t_c}^{t_{\text{end}}} (y(t) - (\hat{\omega}(t; t_c) + C))^2 dt. \quad (98)$$

On writing the cost as an integral. What the estimator evaluates is a sum over uniformly spaced samples, and (98) is its continuous form. Nothing in what follows depends on the two being equal. Every step below — the normal equations, the orthogonality of the residual, Cauchy–Schwarz, and Chebyshev’s inequality — is algebra in an inner product space and holds verbatim whether the inner product is $\int a b dt$ or $T_s \sum_i a_i b_i$; the continuous form is chosen because it makes the moving endpoint and the exchange of integration order in (107) readable, not because the sum is being approximated. The one step that looks continuous-only is the Leibniz cancellation, and its discrete counterpart is exact for the same reason: advancing the onset by one sample transfers that sample from the second segment to the first, where it contributes $(y_j - C)^2$ either way, since $\hat{\omega}(t_c; t_c) = 0$ leaves the two segments equal there.

Only the numerical values of window integrals such as $\|\chi\|^2$ differ, and by a quadrature error. On the $\sinh^2$ integrand the trapezoid relative error is $(2C_2 T_s)^2/12$, which for $C_2 \leq 5.67$ rad/s at $T_s = 10$ ms is below 0.11% — two orders below the smallest term the budget carries.

Three statements elsewhere are *not* available in integral form and are kept discrete throughout: the noise variance $\text{Var}[(Pn)_i] = \sigma_n^2 P_{ii}$, since σ_n is a per-sample deviation and continuous white noise has no pointwise value; the trace and leverage identities $\text{tr } P = 2$ and $P_{ii} \leq 1$, which are statements about a matrix; and the union bound over the N samples of the window. The onset-grid term $T_s \dot{M}/\sqrt{12}$ of Sec. VIII is discrete for the same reason.

Write the recorded rate as the true pre-onset level, the *true* nominal response from the true onset, and a deviation:

$$y(t) = b + \omega_{\text{nom}}(t; t_c^*) + e_\omega(t) + n(t), \quad \hat{t}_c = t_c^* + \delta t_c. \quad (99)$$

Here b is the rate the vehicle actually holds before the excitation and C is its estimate, formed over the pre-onset segment; n is additive gyroscope noise. The onset conditions would put b at zero, and it is worth saying plainly that they do not: measured over the segment the estimator uses, b runs from $0.08^\circ/\text{s}$ at the slowest ramp to 0.71 at the fastest, scaling with $\dot{M}$ at $0.66^\circ/\text{s}$ per Nm/s (correlation 0.58 over 140 runs, $p = 6 \times 10^{-14}$) and carrying the sign of the tip-over in 132 of them. It is not sensor bias — the rate comes from the navigation filter, whose delta-angle bias states remove that — but real motion under a load that is already building. Nor is it constant: the last fifth of the segment runs about twice the first four fifths. Retaining C is therefore not a formality, and Sec. VIII-E reports what pinning it to zero costs. Note also that e_ω is anchored at the *true* onset t_c^* : once a run is recorded it is a fixed function of t , and it carries no dependence on the optimisation variable.

The distinction between ω_{nom} and the model function $\hat{\omega}$ is not cosmetic and must be carried, because the two are not the same curve. $\omega_{\text{nom}} = C_1^* (\cosh C_2^* \tau - 1)$ is what the vehicle does; $\hat{\omega}$ is built from the calibrated constants, $C_1 = K\dot{M}$ and C_2 from stage one, and those carry error. Evaluated at the same true onset the difference does not vanish:

$$\Delta(t) \triangleq \omega_{\text{nom}}(t; t_c^*) - \hat{\omega}(t; t_c^*) = \varepsilon_1 C_1 u(\tau) + O(\varepsilon_2), \quad (99b)$$

with $u(\tau) = \cosh C_2 \tau - 1$, $\varepsilon_1 = C_1^*/C_1 - 1$ and $\varepsilon_2 = C_2^*/C_2 - 1$. Setting Δ to zero — or, equivalently, writing $\hat{\omega}$ in place of ω_{nom} in (99) — would make the data model depend on the estimator’s own output and would

silently charge the calibration error to e_ω , which is defined as the Duhamel integral of ρ and cannot carry it. It is retained below and emerges as a channel of its own.

A first-order expansion of the estimated rate gives

$$\hat{\omega}(t; \hat{t}_c) \simeq \hat{\omega}(t; t_c^*) + \delta t_c \chi(t) \quad (100)$$

with the sensitivity

$$\chi(t) \triangleq \left. \frac{\partial \hat{\omega}(t; t_c)}{\partial t_c} \right|_{t_c=t_c^*} = -C_1 C_2 \sinh C_2 (t - t_c^*). \quad (101)$$

Two features of (98) make the differentiation clean, and both are worth stating because each looks like an omission otherwise.

First, t_c is the upper limit of one integral and the lower limit of the other, so the Leibniz rule produces two boundary terms. They cancel identically. The rise contributes $C_1(\cosh 0 - 1) = 0$ at $t = t_c$, so the two segments agree there and both terms equal $(y(t_c) - C)^2$ with opposite signs. Nothing is discarded, and the cancellation is exact rather than asymptotic.

Second, substituting (99), (99b) and (100), the post-onset residual is

$$y(t) - \hat{\omega}(t; \hat{t}_c) - C = \Delta(t) + e_\omega(t) + n(t) - \delta t_c \chi(t) - (C - b), \quad (102)$$

so C enters only through the baseline *error* $C - b$, not through its own size — a large but correctly estimated pre-onset level is harmless — and the calibration mismatch Δ sits alongside the physical deviation rather than inside it.

Differentiating (98) with respect to δt_c — the same operator as $\partial/\partial t_c$ by (99), but the variable in which the linearised cost is an exact quadratic — and setting the result to zero gives, when the pre-onset fit is unbiased,

$$\frac{\partial J}{\partial \delta t_c} = -2 \int_{t_c}^{t_{\text{end}}} (\Delta + e_\omega + n - \delta t_c \chi - (C - b)) \chi dt = 0, \quad (103)$$

which implies

$$\delta t_c = \frac{\langle e_\omega, \chi \rangle}{\|\chi\|^2} + \frac{\langle \Delta, \chi \rangle}{\|\chi\|^2} + \frac{\langle n, \chi \rangle}{\|\chi\|^2} - (C - b) \frac{\langle 1, \chi \rangle}{\|\chi\|^2}. \quad (104)$$

Here $\langle a, b \rangle$ denotes the L^2 inner product over the fitting window $[t_c^*, t_{\text{end}}]$, i.e. $\langle a, b \rangle \triangleq \int_{t_c^*}^{t_{\text{end}}} a(t)b(t) dt$, and $\|b\|^2 \triangleq \langle b, b \rangle$. The shift δt_c is well defined because the maximum-angle event occurs after the onset, $\tau_{\text{end}} > 0$, so $\sinh C_2 \tau$ does not vanish identically and $\|\chi\|^2 > 0$.

Four channels, and they are of different kinds. The first is the physical deviation and is what (108) bounds a priori. The second is the calibration mismatch of (99b): a *bias*, deterministic once the constants are frozen, and outside anything $\bar{\rho}$ can say. The third is zero in expectation and contributes variance $\sigma_n^2/\|\chi\|^2$. The fourth is the baseline error. Only the first is a statement about the physics the model omits; the rest are statements about what the estimator was given.

The calibration channel. Substituting (99b) with $\Delta = \varepsilon_1 C_1 u$ and using $\langle u, \chi \rangle = -C_1 A(x)$, $A(x) = \frac{1}{4} \cosh 2x - \cosh x + \frac{3}{4}$, the amplitude C_1 cancels twice over and

$$\Delta M_{\text{crit}, \varepsilon} = -\varepsilon_1 \frac{\dot{M}}{C_2} \frac{A(x)}{B(x)} - \varepsilon_2 \frac{\dot{M}}{C_2} \frac{D(x)}{B(x)}, \quad B(x) = \frac{1}{4} \sinh 2x - \frac{1}{2}x, \quad (105a)$$

the second term following the same route for the exponent with $D(x) = \frac{1}{4}x \sinh 2x - \frac{1}{8} \cosh 2x + \frac{1}{8} - \frac{1}{4}x^2$. This is the channel that the two-sided difference removes: $\dot{M}$ changes sign with the tip direction while $\varepsilon_1, \varepsilon_2$ do not, being properties of the configuration, so $\Delta M_{\text{crit}, \varepsilon}$ is odd and cancels in the half-sum $(M_+ + M_-)/2$ that (34) reads. Measured on the campaign, the amplitude mismatch is $\varepsilon_1 = 11.3\%$ and displaces the threshold by 7.2 mN m per direction (24.3 at the fastest ramp) but the reported half-sum by 1.30 mN m, a factor of 5.5 smaller.

The baseline error, retained. The last term of (104) does not vanish either: χ is of one sign, so $\langle 1, \chi \rangle = -C_1(\cosh x - 1)$. Evaluating it with $\|\chi\|^2 = C_1^2 C_2^2 [\frac{1}{4} \sinh 2x - \frac{1}{2}x]$ and carrying it through $\Delta M_{\text{crit}} = \dot{M} \delta t_c$, the amplitude C_1 cancels as it does elsewhere and

$$\Delta M_{\text{crit}, C} = W z_{CoM} (C - b) \frac{\cosh x - 1}{C_2 [\frac{1}{4} \sinh 2x - \frac{1}{2}x]}. \quad (106)$$

Over the operating box of Sec. VI-D the gain of (106) runs from 0.042 N m per rad/s at the slowest ramp to 0.351 at the fastest, the fast end being worse because the window is shorter. What the fitted baselines actually are, and how far the identified offset moves when C is pinned to zero instead of fitted, are measurements and are reported with the realised budget in Sec. VIII-E.

From the onset shift to the threshold. What follows treats the *first* term of (104) only; the other three have just been accounted for and are added to it, not absorbed by it. Substituting the Duhamel integral (88) for e_ω in (104) and exchanging the order of integration — Fubini, both integrands being continuous on a bounded domain, the inner limits changing from $s \in [0, \tau]$ to $\tau \in [s, \tau_{\text{end}}]$ — turns the correlation into an average of ρ against a fixed weight:

$$\Delta M_{\text{crit},e} = - \int_0^{\tau_{\text{end}}} \rho(s) w(s) ds, \quad w(s) = \frac{Wz_{CoM} \int_s^{\tau_{\text{end}}} \sinh(C_2\tau) \cosh(C_2(\tau-s)) d\tau}{J_P C_2 \|\sinh(C_2\tau)\|^2}. \quad (107)$$

Three properties of w close the argument. It is non-negative, being an integral of hyperbolic functions of non-negative arguments. It integrates to one *exactly*: the same exchange over the triangle $0 \leq s \leq \tau \leq \tau_{\text{end}}$ reproduces in the numerator the very norm that sits in the denominator, leaving $Wz_{CoM}/(J_P C_2^2) = 1$ by (20). And it is non-increasing in s , since both the integration range and the cosh factor shrink as s grows.

The first two properties alone give $|\Delta M_{\text{crit},e}| \leq \langle |\rho| \rangle_w$, the w -weighted average of $|\rho|$, since $w \geq 0$ and $\int w = 1$. The third admits Chebyshev's inequality — a forcing that rises against a weight that falls, the same pairing as before — and replaces that average by the one already bounded at (95).

It must be applied to the two channels *separately*, and the reason is not cosmetic: ρ itself does not rise. Near the onset $\varphi \sim \tau^3$ makes $\rho_\varphi \propto \varphi^2$ of order τ^6 while $\rho_{GE} \propto \dot{M}\tau\varphi$ is of order τ^4 , so the ground-effect term dominates there; being restoring, it holds ρ *negative* over the first 13 to 24% of the window (integrating the exact nonlinear tip-over, Sec. VIII). Taken on their own the two channels are monotone throughout: $\rho_\varphi \geq 0$ rises because φ does, and $|\rho_{GE}| = |\beta_M| \dot{M}\tau\varphi$ rises for the same reason. Applying the inequality to each and adding,

$$|\Delta M_{\text{crit},e}| \leq \langle \rho_\varphi \rangle_w + \langle |\rho_{GE}| \rangle_w \leq \bar{\rho} \leq \frac{1}{7} \rho_{\varphi,\text{max}} + \frac{1}{5} \rho_{GE,\text{max}}. \quad (108)$$

The two reduction factors are therefore produced by the step that uses them, rather than imported past a hypothesis the sum does not satisfy. Note that $\bar{\rho}$ is the average defined at (95), and the right-hand side bounds it rather than defining it. The constants are the reduction factors themselves, not the factors multiplied by a kernel gain, because the onset is located by a *normalised* correlation: $\int w = 1$ leaves nothing over.

That each channel rises across the window is what makes the Chebyshev step available, and it should be read as a hypothesis that happens to hold rather than as a conservatism. The weight is *largest at the onset*: $w(0) = C_2 \sinh^2 x / 2 [\frac{1}{4} \sinh 2x - \frac{1}{2}x]$, which is 5.16 s^{-1} at $\dot{M} = 1.20 \text{ N m/s}$ against a mean of 1.63, and it falls monotonically from there. Forcing that acts early is therefore weighted *most*, not least, and it is only because $\rho_\varphi \propto \varphi^2$ and $\rho_{GE} \propto \tau\varphi$ both vanish at the onset and rise afterwards that the pairing runs the favourable way. Reversing the arrangement reverses the conclusion: holding the window mean fixed at $\bar{\rho}$ and letting ρ *fall* across the window instead, the identified threshold moves by 1.4 to 1.6 $\bar{\rho}$, outside (108). The monotonicity is doing real work and is not a technicality.

It is tempting to argue instead that an early perturbation cannot displace the onset at all, on the grounds that expanding the kernel of (88) as $\cosh C_2(\tau-s) = \cosh C_2\tau \cosh C_2s - \sinh C_2\tau \sinh C_2s$ leaves the $\sinh$ half weighted by $\sinh C_2s$, which vanishes at $s = 0$. That reasoning assigns the cosh half to the amplitude, and the amplitude is pinned, so it is not available to take it. What the cosh half does instead is project onto the onset direction, and it projects strongly: over this window $\cosh C_2\tau$ and $\sinh C_2\tau$ are separated by only 3.4° once the constant is removed, so 99.8% of an amplitude perturbation lies along the directions the estimator can move in. A pure 2% amplitude error, with no onset error whatever, displaces the identified threshold by 4.3 mN m — a third of the budget of (108) — while changing the fit residual by $0.024^\circ/\text{s}$. Equation (107) already accounts for this, being derived from the projection rather than from matching coefficients, which is why $w(0)$ comes out as the maximum.

The constants differ between the two halves of (97) because the two questions do. The first asks how far the fitted rate curve is displaced, so the kernel integral Ψ multiplies $\bar{\rho}$ and the products climb to $\frac{1}{2}$ and 1; the second asks how far the *minimiser* moves, and by (107) that is a normalised correlation with no kernel gain left over, so $\bar{\rho}$ enters bare and the constants are the reduction factors themselves. The threshold bound is the stronger of the two by about a factor of four, which is why it, and not the relative rate bound, is what the offset budget inherits. A reviewer can evaluate either from numbers already in the paper.

Fit quality and onset accuracy are separate claims. Two things are easily merged here and should not be. One is whether the hyperbolic cosine can *fit* the recorded rate; the other is whether the onset it selects is the right one. Neither implies the other, and they are established by different arguments.

The fitting claim is conditional on an onset and holds at every instant. Its parametrisation satisfies the onset conditions identically for *any* candidate t_c , since $\cosh 0 - 1 = 0$ and $\sinh 0 = 0$ give $\hat{\omega}(t_c; t_c) = 0$ and $\hat{\omega}(t_c; t_c) = 0$, so no admissible fit can violate the physics at the onset whichever one is selected. Given that onset, the two remaining parameters enter linearly, the minimiser is the orthogonal projector P onto their span, and the nominal — lying in that span — cancels from the residual exactly, leaving $r = (I - P)(e_\omega + n)$ and

$$|r_i| \leq E_i + \sum_j |P_{ij}| E_j + k\sigma_n(1 + \sqrt{P_{ii}}), \quad (109)$$

which every sample of all 140 runs satisfies, the worst reaching 0.763 of it. That is a statement about fit quality and about nothing else.

The accuracy of the onset is (108), and its evidence lies at the other end of the window. The sensitivity χ of (101) vanishes at the onset, so the record there is nearly uninformative about the quantity being located: with $\|\chi\|^2 = C_1^2 C_2 [\frac{1}{4} \sinh 2x - \frac{1}{2}x]$, the final quarter of the window carries 70% of that norm at $x = 1.98$ and 84% at $x = 3.65$, while the first quarter carries 0.8% and 0.2%. The onset is fixed by the far end of the rise, not by the neighbourhood of the onset itself, and the residual being small near t_c is not evidence that t_c is right.

This is worth stating explicitly because the campaign contains the corresponding failure. At $\dot{M} = 0.10$ N m/s the identified threshold reads 26.70 mN m low with the sign of an onset found *early*, which is what a weak rise emerging from noise does to a minimiser whose informative samples have not yet arrived. It also points the same way as the recommendation on excitation rate: $\|\chi\|$ grows with $C_1 = \dot{M}/Wz_{CoM}$, so the quantity that locates the onset improves with a faster ramp even as the window shortens.

What (108) covers. Only the first term of (104), the displacement driven by the forcing the estimator cannot absorb. Collecting the rest,

$$|\Delta M_{\text{crit}}| \leq \underbrace{\bar{\rho}}_{(108)} + \underbrace{|\Delta M_{\text{crit},\varepsilon}|}_{(105a)} + \underbrace{|\Delta M_{\text{crit},C}|}_{(106)} + \dot{M} \frac{|\langle n, \chi \rangle|}{\|\chi\|^2}, \quad (110)$$

of which only the first is bounded a priori from the geometry. The second and third are properties of the calibration and of the pre-onset segment. Their realised sizes are not in the order the bound suggests: the $\bar{\rho}$ channel is allowed 11.5 mN m but realises 0.3 to 1.3, the baseline channel realises 0.02 to 0.18, and the calibration channel realises 7.2 per direction — five times the other two together, and the only one of the three the two-sided half-sum removes, taking it to 1.30. A fourth contribution sits outside (110) altogether: the applied moment departs from the linear ramp the model assumes, and because $\dot{M}$ is the least-squares slope of the *measured* moment over the window, the constant and linear parts of any tracking error are removed by the normal equations while the non-affine remainder survives and pairs with the non-affine part of w . All of these are reported with the realised budget in Sec. VIII-E, alongside the ground-contact geometry that dominates them.